MIL-W-63451 (AR)
17 JUNE 1981

MILITARY SPECIFICATION

WARHEAD, GUIDED MISSILE, HE, M207E2
LINER FOR

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings for the fabrication and packing of liner for M207E2 Warhead.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9340752 Liner

FSC: 1336

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawing 9340752, all associated drawings and with requirements of applicable standards and specifications.

3.3 Data requirements. The contractor shall generate data in accordance with requirements of data item description cited in 6.4.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. All parts shall be fabricated in a thorough, workmanlike manner. The parts shall be free of burrs, chips, sharp edges, cracks, unblended radii, dents, tool marks, chipped edges, warpage, gouges, surface defects, dirt, grease, corrosion products, burn marks, and other foreign matter. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detailed specification.

4.2 Classification of inspections. The following types of inspections shall be conducted.

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection

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4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following item in sample quantities as indicated.

Part Description	Drawing	Quantity
Liner	9340752	20

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Liner		SHEET 1 OF 1		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Liner, (Dwg. 9340752)				
	Examination for Defects	20		3.2	4.4.2.1

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection.

4.4.2 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of warhead metal parts shall contain:

a. Liner 9340752 from not more than one interfix lot number from one manufacturer.

4.4.2 Examinations. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Tests tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Liner				9340752
					NEXT HIGHER ASSEMBLY
					9340753
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Runout of O.D. with datums A & B		0.40%	3.2	Gage
102	Outside diameter (O. D.)		0.40%	3.2	Gage
103	Inside diameter (I.D.)		0.40%	3.2	Gage
104	Length of liner step		0.40%	3.2	Gage
105	Length from base to datum B		0.40%	3.2	Gage
106	Straightness on large outside surface		0.25%	3.2	Gage
107	Runout of base with datums A & B		0.25%	3.2	Gage
108	Length from datum A to B		0.40%	3.2	Gage
109	Length from datum B to inside spherical radius		0.40%	3.2	Gage
110	Wall thickness at large datum, 42° section		0.40%	3.2	Gage
111	Wall thickness at small datum, 42° section		0.40%	3.2	Gage
112	Wall thickness at apex end		0.40%	3.2	Gage
113	Wall thickness at large basic dimension at apex end		0.25%	3.2	Gage
114	Wall thickness at small basic dimension, apex end		0.25%	3.2	Gage
115	Length from large to small basic dimensions at apex end		0.40%	3.2	Gage
116	Runout between two basic dimensions on outside surfaces at apex end (See Note 8 on dwg.)		0.25%	3.2	Gage
117	Length from datum B to basic dimension at apex end		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Liner	2 OF 2	9340752
			NEXT HIGHER ASSEMBLY 9340753

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MAJOR (Continued)					
118	Wall thickness variation in any axial plane		0.25%	3.2	Gage
119	Wall thickness variation in any transverse plane		0.25%	3.2	Gage
MINOR					
201	Surface finish improper		0.65%	3.2	Gage
202	Inside spherical radius, apex end (See Note 1)			3.2	Gage
203	Outside spherical radius, apex end		0.65%	3.2	Gage
204	Presence of 4 slots on O.D.		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.00%	3.5	Visual

NOTE: 1. All new and resharpened tooling shall be certified prior to use. A cast of the inside spherical radius of the liner shall be made from "dental plaster" or equivalent, which shall be inspected by an optical comparator of 10X magnification or greater. The inspection shall be performed at the beginning of the shift, half way through the shift and whenever tooling is changed.

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4.4.3 Testing. Testing is described in First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein and Table 1.

5. PACKAGING

5.1 Packaging and packing. The parts shall be packaged and packed in accordance with the requirements of MIL-STD-1169.

5.2 Marking. Marking shall be in accordance with the requirements of MIL-STD-1169.

6. NOTES

6.1 Intended use. The liner covered by this specification is a component of the Improved TOW Warhead Section, Guided Missile, HE, M207E2.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

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6.4 Data requirements. Deliverable data required by the specification cited as follows:

Data Requirement	Applicable Data Item
Ammunition Data	DI-L-1410

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1336-A400

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-W-63451 WARHEAD, GM. HE, M207E2 LINER FOR

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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